

Unified Smart Laboratory Ecosystem

SECTION A: Respondent Information

Laboratory Technician Questionnaire

Please provide the following background information. This section is used for demographic analysis only and will not be used to identify individual respondents.

Jomo Kenyatta University of Agriculture and Technology (JKUAT)

Q1. What is your current role in the laboratory?

- ☐ Laboratory Technician
- ☐ Senior Laboratory Technician
- ☐ Laboratory Superintendent
- ☐ Research Assistant
- ☐ Other (specify): _____

Research Study: IoT-Based Smart Laboratory Systems in Higher Education

Q2. Which department/laboratory do you work in? (Tick all that apply)

- ☐ Physics Laboratory
- ☐ Chemistry Laboratory
- ☐ Engineering Laboratory
- ☐ Clinical/Medical Laboratory
- ☐ Computer Laboratory
- ☐ Biology Laboratory

INSTRUCTIONS TO RESPONDENTS

- This questionnaire is for research purposes only. All responses are strictly confidential.
- Please answer all questions as accurately and honestly as possible.
- For multiple choice questions, tick (■) all options that apply unless stated otherwise.
- For open-ended questions, write legibly in the space provided.
- The questionnaire should take approximately 20–25 minutes to complete.

Q3. How many years have you worked in laboratory management?

- ☐ Less than 1 year
- ☐ 1 – 3 years
- ☐ 4 – 6 years
- ☐ 7 – 10 years
- ☐ More than 10 years

Q4. What is your highest level of education?

- ☐ Certificate
- ☐ Diploma

- ☐ Bachelor's Degree
- ☐ Master's Degree
- ☐ Other: _____

By completing this form, you consent to your responses being used for academic research.

Q5. Have you previously used any digital or computer-based laboratory management system?

- ☐ Yes ☐ No ☐ Not Sure

If yes / explain:

SECTION B: Current Laboratory Practices

This section evaluates the current methods used to manage laboratory operations at your institution.

Q6. How is student attendance currently recorded in your laboratory? (Select all that apply)

- ☐ Physical sign-in register
- ☐ Lecturer/technician verbal roll call
- ☐ Student identity card scan
- ☐ Digital system (please specify below)
- ☐ No formal record is kept

Q7. How do you currently track laboratory equipment usage?

- ☐ Manual logbook
- ☐ Spreadsheet (Excel/Google Sheets)
- ☐ Dedicated software system
- ☐ Equipment is not formally tracked
- ☐ Other: _____

Q8. How are practical sessions currently documented?

- ☐ Students write reports in physical lab notebooks
- ☐ Printed report templates submitted on paper
- ☐ Digital reports submitted via email or portal
- ☐ No formal documentation system
- ☐ Other: _____

Q9. What are the most significant challenges you face in managing laboratory sessions?

(Open-ended response)

Note: Please be as specific as possible. Examples: missing equipment, student disputes, poor record keeping.

Q10. On a scale of 1 to 5, how would you rate the overall efficiency of your current laboratory management system?

1 Strongly Disagree	2 Disagree	3 Neutral	4 Agree	5 Strongly Agree
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Please circle your response above.

Q11. How frequently do issues arise due to poor or incomplete documentation of practical sessions?

- ☐ Never
- ☐ Rarely (once per semester)
- ☐ Sometimes (monthly)

■ Often (weekly)

■ Very often (daily or almost daily)

Q12. How are disputes about student attendance or practical completion currently resolved?

(Open-ended response)

SECTION C: Awareness and Initial Perception of the Proposed USLE

The Unified Smart Laboratory Ecosystem (USLE) is a proposed integrated digital platform combining biometric authentication, IoT-based equipment tracking, digital lab notebooks, technician approval workflows, and analytics. Please respond based on this description.

Q13. Before reading the above description, were you aware of any similar smart laboratory management systems being used locally or internationally?

- ☐ Yes, I am aware of several such systems
- ☐ Yes, I have heard of one or two
- ☐ No, I was not aware of any
- ☐ I have used such a system before

Q14. After reviewing the USLE description, how would you rate its relevance to the needs of your laboratory?

1 Strongly Disagree	2 Disagree	3 Neutral	4 Agree	5 Strongly Agree
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Please circle your response above.

Q15. Which of the following proposed USLE features do you find most useful? (Tick all that apply)

- ☐ Biometric fingerprint authentication
- ☐ RFID card fallback authentication
- ☐ Staff-issued time-limited QR code (emergency fallback)
- ☐ IoT-based equipment tracking
- ☐ Digital lab notebook
- ☐ Group submission with individual verified copies
- ☐ Technician digital approval and signature
- ☐ Analytics and performance dashboards
- ☐ Cross-lab access for students
- ☐ Integration with Clinical/LIMS systems

Q16. What concerns or reservations do you have about implementing a system like USLE in your laboratory?

(Open-ended response)

Q17. How strongly do you agree that a system like USLE would address the challenges you described in Section B?

1 Strongly Disagree	2 Disagree	3 Neutral	4 Agree	5 Strongly Agree
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Please circle your response above.

SECTION D: Authentication System Feedback

Q18. How reliable do you consider your current method of verifying student identity at the start of a practical session?

1 Strongly Disagree	2 Disagree	3 Neutral	4 Agree	5 Strongly Agree
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Please circle your response above.

Q19. Do you believe biometric fingerprint authentication would improve student accountability in your laboratory?

☐ Yes ☐ No ☐ Not Sure

If yes / explain:

Q20. How practical do you think it is to enrol all students in a fingerprint recognition system at the start of each academic year?

1 Strongly Disagree	2 Disagree	3 Neutral	4 Agree	5 Strongly Agree
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Please circle your response above.

Q21. On a scale of 1 to 5, how useful would a technician-controlled, time-limited QR code be as a fallback when biometric authentication is unavailable?

1 Strongly Disagree	2 Disagree	3 Neutral	4 Agree	5 Strongly Agree
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Please circle your response above.

Q22. How concerned are you about the possibility of students bypassing or misusing the authentication system?

- ☐ Not concerned at all
- ☐ Slightly concerned
- ☐ Moderately concerned
- ☐ Very concerned
- ☐ Extremely concerned

Q23. Do you foresee any specific technical or operational challenges with implementing a multi-layer authentication system?

(Open-ended response)

Q24. How important is it that the system can operate offline during network outages, with data synced later?

1 Strongly Disagree	2 Disagree	3 Neutral	4 Agree	5 Strongly Agree
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Please circle your response above.

SECTION E: Digital Lab Notebook and Reporting

Q25. How are student experimental observations and results currently recorded during practical sessions?

- ☐ Handwritten in physical lab notebooks
- ☐ On printed lab sheet templates
- ☐ Verbally recorded and written up later
- ☐ No standard method — varies by student
- ☐ Other: _____

Q26. On a scale of 1 to 5, how much do you think a structured digital lab notebook would improve the quality and completeness of student records?

1 Strongly Disagree	2 Disagree	3 Neutral	4 Agree	5 Strongly Agree
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Please circle your response above.

Q27. How practical do you consider it for students to use laptops or tablets to record data during laboratory experiments?

1 Strongly Disagree	2 Disagree	3 Neutral	4 Agree	5 Strongly Agree
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Please circle your response above.

Note: Consider factors such as available space, power sockets, and type of practical.

Q28. Would you support a group submission model where one student submits a practical report on behalf of the group and the system automatically distributes verified individual copies to all group members?

- ☐ Yes ☐ No ☐ Not Sure

If yes / explain:

Q29. How confident are you that a digital lab notebook would reduce cases of copied, forged, or plagiarised practical reports?

1 Strongly Disagree	2 Disagree	3 Neutral	4 Agree	5 Strongly Agree
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Please circle your response above.

Q30. What specific features should a digital lab notebook include to best meet the needs of your laboratory?

(Open-ended response)

SECTION F: Technician Approval and Digital Signatures

Q31. How do you currently verify and confirm that a student has successfully completed a practical session?

(Open-ended response)

Q32. On a scale of 1 to 5, how much would a formal technician digital approval step improve accountability in the submission process?

1 Strongly Disagree	2 Disagree	3 Neutral	4 Agree	5 Strongly Agree
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Please circle your response above.

Q33. How valuable do you consider a cryptographically signed digital lab report — one that cannot be altered after approval — for assessment and audit purposes?

1 Strongly Disagree	2 Disagree	3 Neutral	4 Agree	5 Strongly Agree
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Please circle your response above.

Q34. How manageable do you think it would be to review and digitally approve student submissions, given your current workload?

- ☐ Very manageable — I have adequate time
- ☐ Manageable with minor workflow adjustments
- ☐ Challenging but possible with support
- ☐ Very challenging — would require additional staffing
- ☐ Not feasible given current workload

Q35. Do you foresee any challenges in reviewing and approving digital practical submissions?

(Open-ended response)

Q36. Would you prefer to approve submissions via a dedicated desktop portal, a mobile application, or both?

- ☐ Desktop portal only
- ☐ Mobile application only

- Both desktop and mobile
- No preference

SECTION G: Lab-Specific Questions

Please answer *ONLY* the subsection(s) that correspond to the laboratory/laboratories you manage. You may skip subsections that are not relevant to your role.

G1 — Physics Laboratories

Q37. What are the most common types of equipment used in your physics practicals? (Tick all that apply)

- ☐ Oscilloscopes
- ☐ Function generators
- ☐ Optical benches
- ☐ Vernier callipers / micrometers
- ☐ Power supplies
- ☐ Motion sensors / data loggers
- ☐ Other: _____

Q38. What specific challenges exist in recording and verifying experimental readings during physics practicals?

(Open-ended response)

Q39. Would real-time digital recording of experimental readings improve accuracy and reduce transcription errors?

1 Strongly Disagree	2 Disagree	3 Neutral	4 Agree	5 Strongly Agree
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Please circle your response above.

G2 — Chemistry Laboratories

Q40. How are chemicals currently tracked and managed in your laboratory? (Select all that apply)

- ☐ Manual chemical inventory register
- ☐ Dedicated chemical management software
- ☐ Spreadsheet (Excel/Google Sheets)
- ☐ Informal tracking by technician memory
- ☐ Other: _____

Q41. What safety monitoring systems are currently in place in your chemistry laboratory?

- ☐ Gas leak detectors
- ☐ Automated fume hood monitoring
- ☐ Manual safety checks by technician
- ☐ Temperature/humidity sensors
- ☐ No formal monitoring system
- ☐ Other: _____

Q42. Would IoT-based real-time safety monitoring (e.g., gas, temperature, humidity sensors) improve laboratory safety outcomes?

1 Strongly Disagree	2 Disagree	3 Neutral	4 Agree	5 Strongly Agree
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Please circle your response above.

G3 — Engineering Laboratories

Q43. What types of equipment and machines are available in your engineering laboratory? (Tick all that apply)

- ☐ CNC machines / lathes
- ☐ Oscilloscopes / signal analysers
- ☐ 3D printers
- ☐ Welding equipment
- ☐ Electronic workstations
- ☐ Hydraulic/pneumatic rigs
- ☐ Other: _____

Q44. How is equipment usage currently scheduled for students?

- ☐ First-come, first-served on the day
- ☐ Advance booking via physical register
- ☐ Timetabled sessions managed by technician
- ☐ Online booking system
- ☐ No formal scheduling — informal arrangement

Q45. Would a digital equipment tracking and scheduling system improve utilisation rates and reduce student waiting times?

1 Strongly Disagree	2 Disagree	3 Neutral	4 Agree	5 Strongly Agree
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Please circle your response above.

G4 — Clinical / Medical Laboratories

Q46. How are patient samples currently tracked and documented in your laboratory?

(Open-ended response)

Q47. How is chain-of-custody currently maintained for samples handled by students?

- ☐ Physical chain-of-custody forms signed at each transfer point
- ☐ Electronic LIMS system
- ☐ Informal verbal hand-offs

- No formal chain-of-custody system
- Other: _____

Q48. Would a digital chain-of-custody system integrated with USLE improve the reliability and auditability of sample handling?

1 Strongly Disagree	2 Disagree	3 Neutral	4 Agree	5 Strongly Agree
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Please circle your response above.

Q49. What is your greatest concern regarding student access to patient samples and clinical data?
(Open-ended response)

SECTION H: System Flexibility and Inter-Lab Collaboration

Q50. How often do students need to redo or reschedule practical sessions at your laboratory?

- ☐ Rarely (a few per semester)
- ☐ Occasionally (monthly)
- ☐ Frequently (weekly)
- ☐ Very frequently (daily)

Q51. Would allowing students from other departments to access your laboratory (with appropriate authorisation) improve overall practical completion rates?

1 Strongly Disagree	2 Disagree	3 Neutral	4 Agree	5 Strongly Agree
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Please circle your response above.

Q52. How useful would inter-laboratory equipment sharing (e.g., Physics using Chemistry sensors) be for your institution?

1 Strongly Disagree	2 Disagree	3 Neutral	4 Agree	5 Strongly Agree
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Please circle your response above.

Q53. Do you support the idea of USLE being extended to support collaboration with external/private laboratories?

- ☐ Yes ☐ No ☐ Not Sure

If yes / explain:

Note: Examples: industry research labs, hospital laboratories, private testing facilities.

Q54. What safeguards or conditions would you require before supporting external laboratory collaboration?
(Open-ended response)

SECTION I: Implementation and Technology Adoption

Q55. On a scale of 1 to 5, how ready do you feel your laboratory is to adopt a digital management system like USLE?

1 Strongly Disagree	2 Disagree	3 Neutral	4 Agree	5 Strongly Agree
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Please circle your response above.

Q56. What type and level of training would you require before using the USLE system effectively?

- ☐ Basic one-day orientation session
- ☐ Multi-day hands-on workshop
- ☐ Ongoing support and refresher training
- ☐ Self-paced online training materials
- ☐ Peer-to-peer training from early adopters
- ☐ I would not require training

Q57. What do you consider to be the most significant barriers to implementing USLE at JKUAT?

- ☐ High initial cost of hardware (fingerprint scanners, RFID readers, etc.)
- ☐ Unreliable internet connectivity
- ☐ Resistance to change from staff or students
- ☐ Insufficient technical support for maintenance
- ☐ Privacy and data security concerns
- ☐ Lack of institutional policy support
- ☐ Other: _____

Q58. In your opinion, what additional infrastructure would need to be in place before USLE could be successfully deployed?

(Open-ended response)

Q59. How strongly do you agree that the benefits of USLE would outweigh the costs and effort of implementation?

1 Strongly Disagree	2 Disagree	3 Neutral	4 Agree	5 Strongly Agree
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Please circle your response above.

SECTION J: Final Feedback and Recommendations

Q60. What specific improvements or additional features would you suggest for the USLE system to better serve your laboratory?
(Open-ended response)

Q61. Overall, do you believe the Unified Smart Laboratory Ecosystem would improve laboratory management and student outcomes at JKUAT?
☐ Yes ☐ No ☐ Not Sure
If yes / explain:

Q62. On a scale of 1 to 5, how likely are you to personally adopt and champion the use of USLE in your laboratory if it were introduced?

1 Strongly Disagree	2 Disagree	3 Neutral	4 Agree	5 Strongly Agree
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Please circle your response above.

Q63. Any additional comments, observations, or recommendations for the research team?
(Open-ended response)

Thank you for your time and valuable input. Your responses will directly contribute to the design and development of the Unified Smart Laboratory Ecosystem at JKUAT. For queries regarding this research, please contact the principal investigator through the Department of Information Technology.